

Organic chemistry

4CHM311

**Learning Unit 4: REACTIONS AT THE ALPHA (α) CARBON
OF CARBONYL COMPOUNDS (Enols and Enolates)**

Ch. 18 & 19

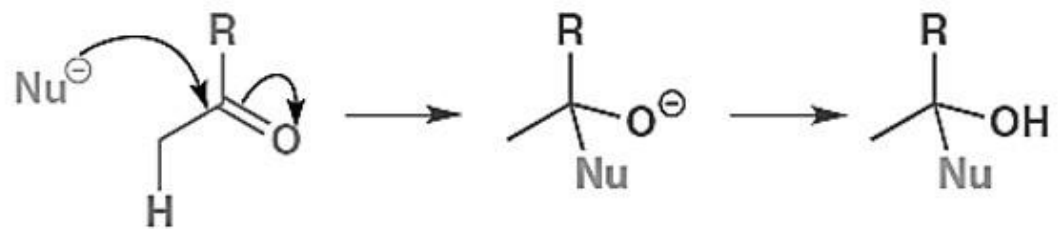
Reactions at the α Carbon of Carbonyl Compounds

Learning Objectives

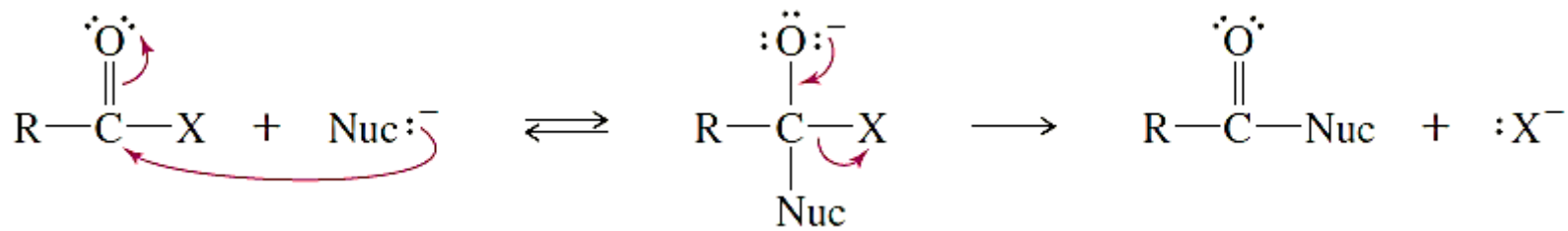
1. Show how enols and enolate ions act as nucleophiles
2. Predict the products of their reactions with alkyl halides, and other electrophiles
3. Show how they are useful in synthesis
4. Predict the products of aldol condensations, and draw mechanisms to show how they are formed
5. Show how these condensations are useful in synthesis

- Carbonyl compound acts as an electrophile in nucleophilic addition and nucleophilic acyl substitution

Nucleophilic addition

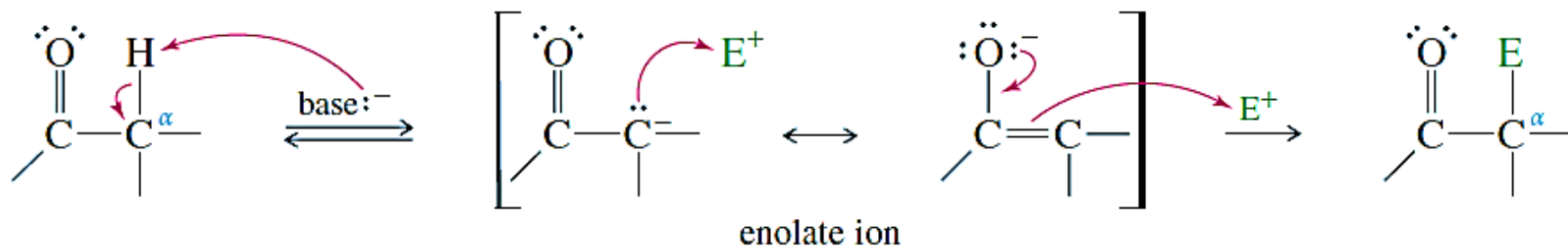


Nucleophilic acyl substitution

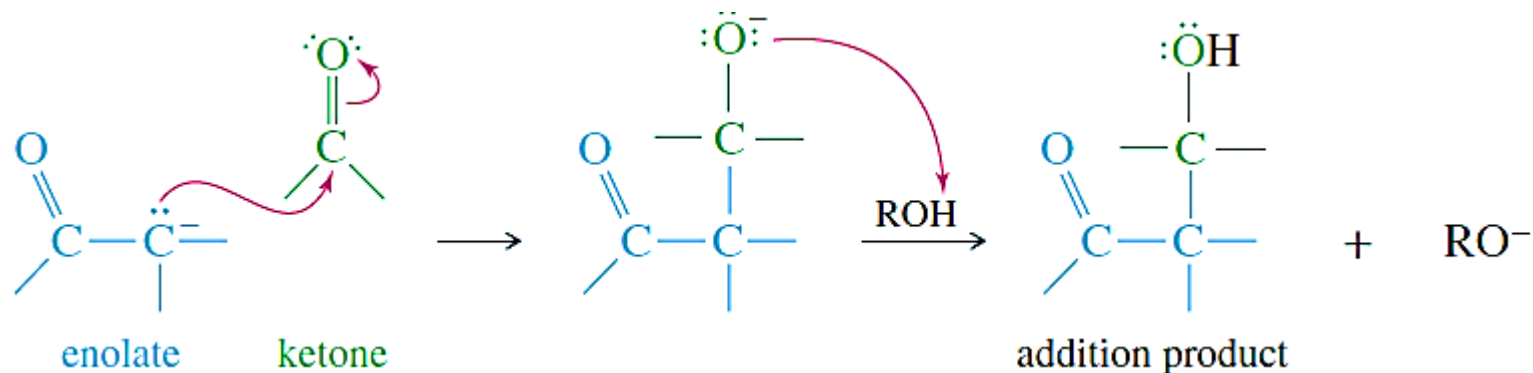


Alpha Substitutions and Condensations of Carbonyl Compounds

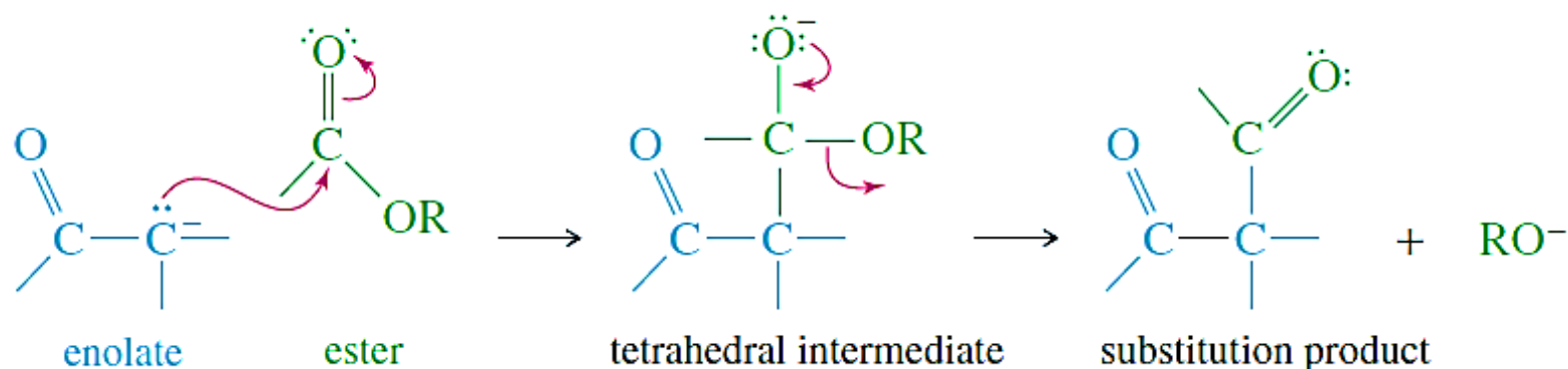
- Carbonyl compound acts as a nucleophile
- **Alpha substitution:** Substitution at the carbon atom next to the carbonyl group
- Alpha (α) substitutions involve the replacement of a hydrogen atom at the α carbon atom by some other group



- **Carbonyl condensations:** alpha substitutions where the electrophile is another carbonyl compound
- If the electrophile is a ketone or an aldehyde, the enolate ion adds to that carbonyl group in a nucleophilic addition mechanism



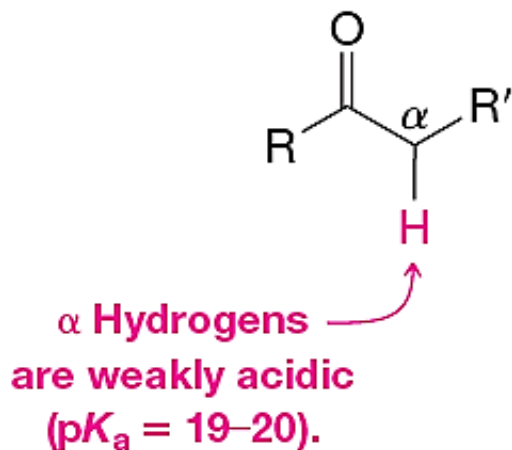
- If the electrophile is an ester, the ester undergoes a nucleophilic acyl substitution



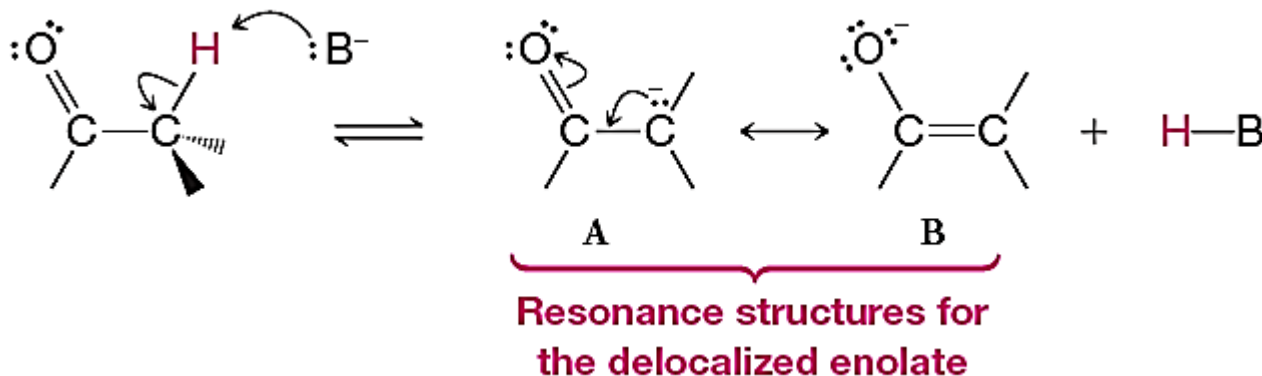
- Alpha substitutions and condensations of carbonyl compounds are some of the most common methods for forming carbon-carbon bonds

Enols, Enolates and Enamines

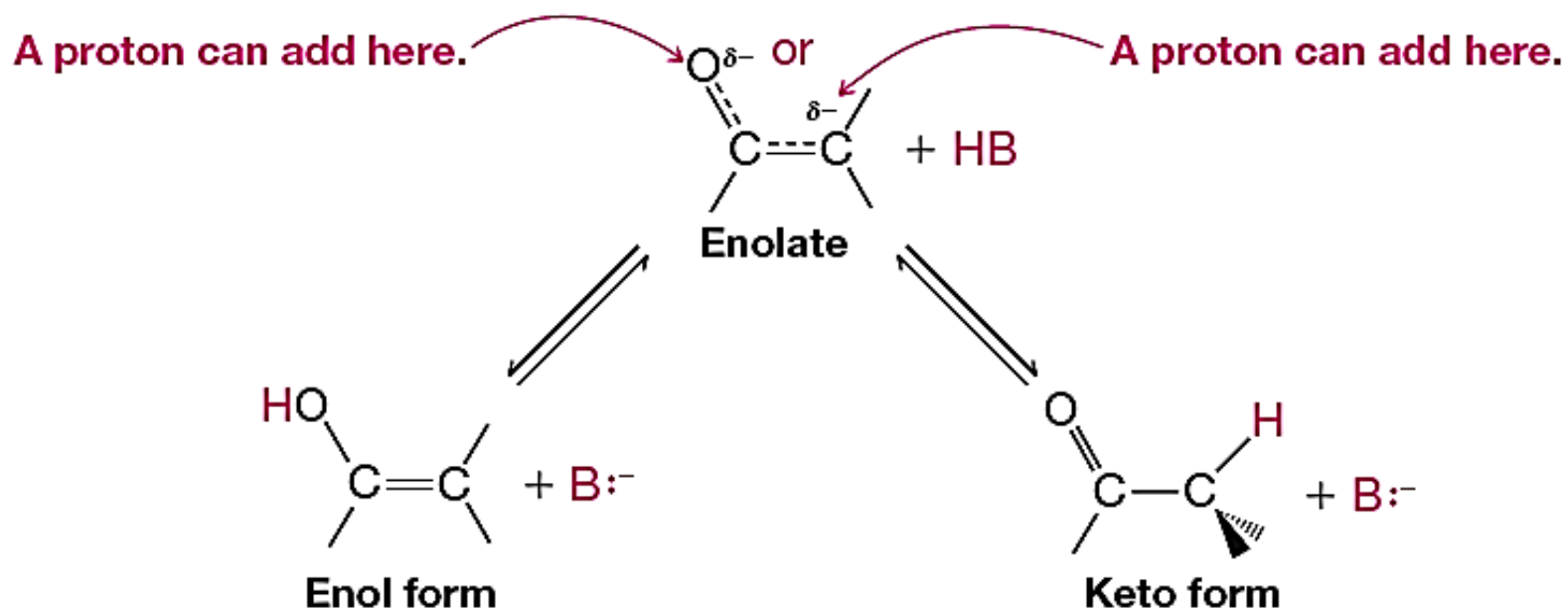
- The α hydrogens of carbonyl compounds are acidic



- When a carbonyl compound loses an α proton, the anion that is produced, called an **enolate**, is stabilized by delocalization

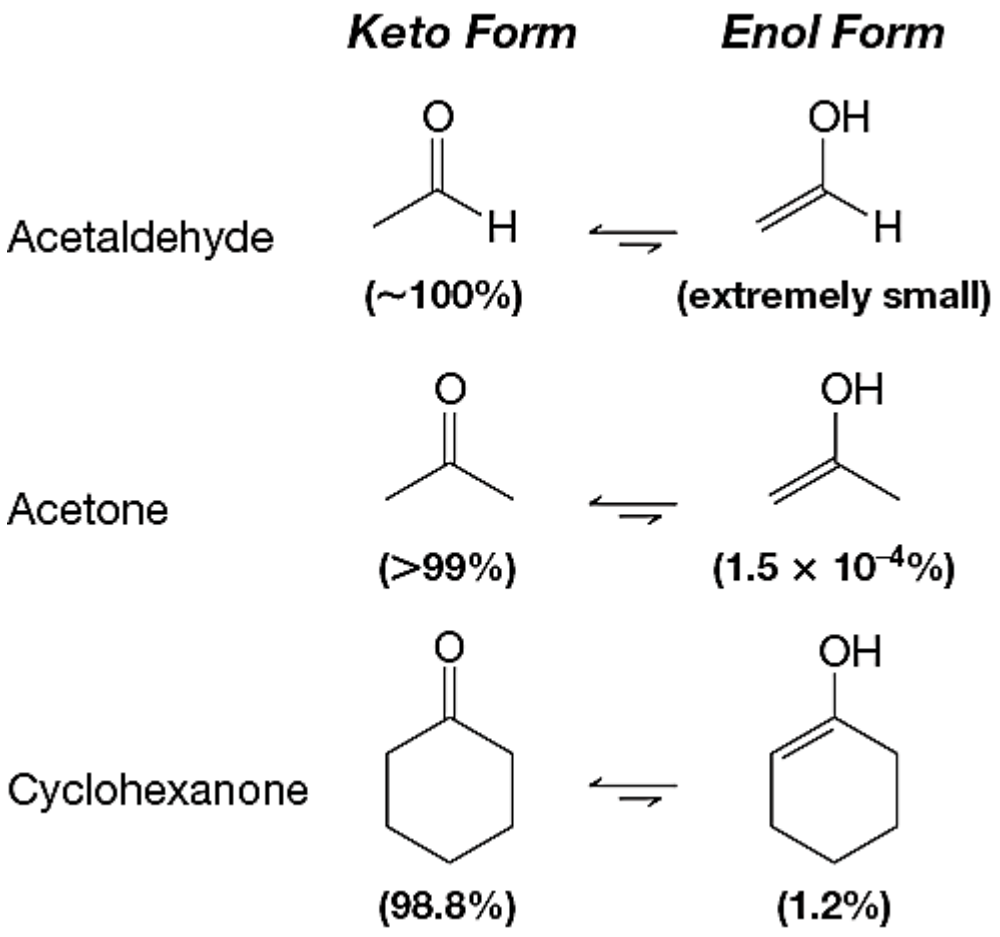


- The resonance-stabilized enolate can accept the proton at carbon to form the original carbonyl compound (the **keto form**) or it may accept the proton at oxygen to form an **enol** (alkene alcohol)

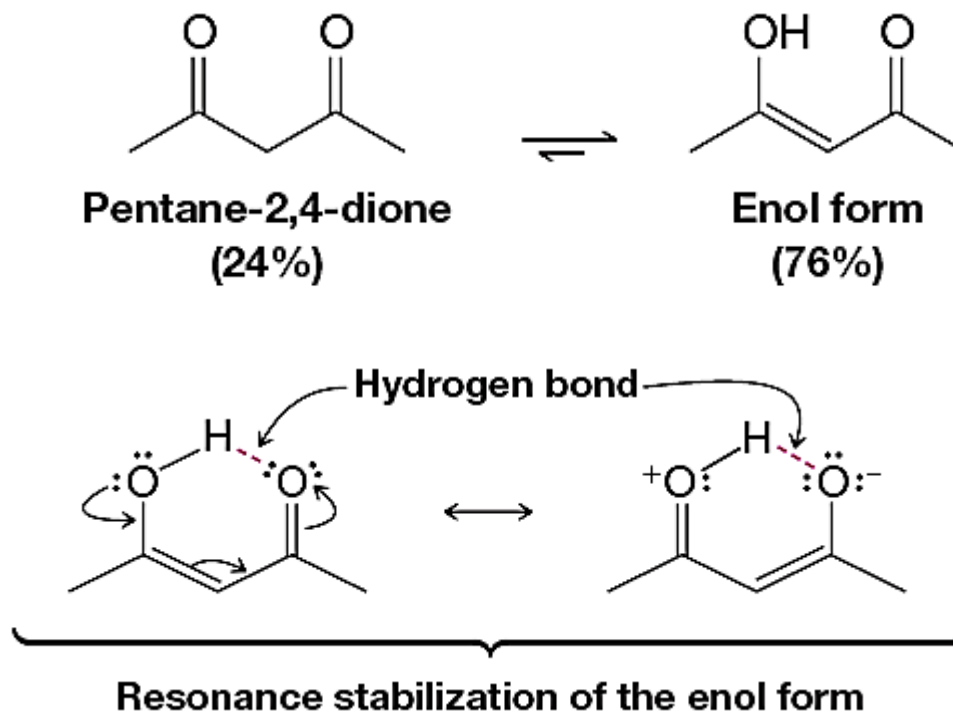


Keto and Enol Tautomers

- The constitutional isomers **keto** and **enol** forms of carbonyl compounds are **tautomers**, and their interconversions called **tautomerization**.



- A molecule with two carbonyl groups separated by one carbon atom (called β -dicarbonyl compounds) has higher amount of enol at equilibrium

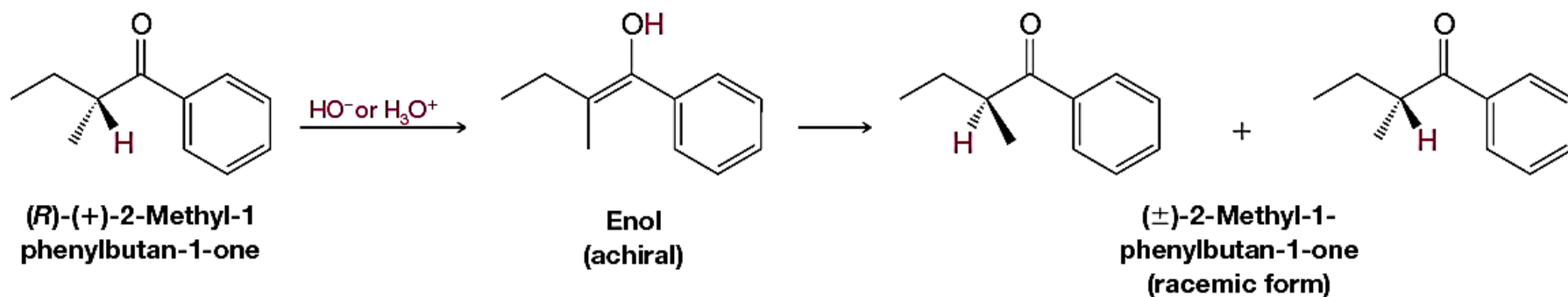


Exercise: The compound cyclohexa-2,4-dien-1-one exists totally in its enol form. Write the structure of cyclohexa-2,4-dien-1-one and of its enol form. What special factor accounts for the stability of the enol form?

Reactions via Enols and Enolates

Racemization

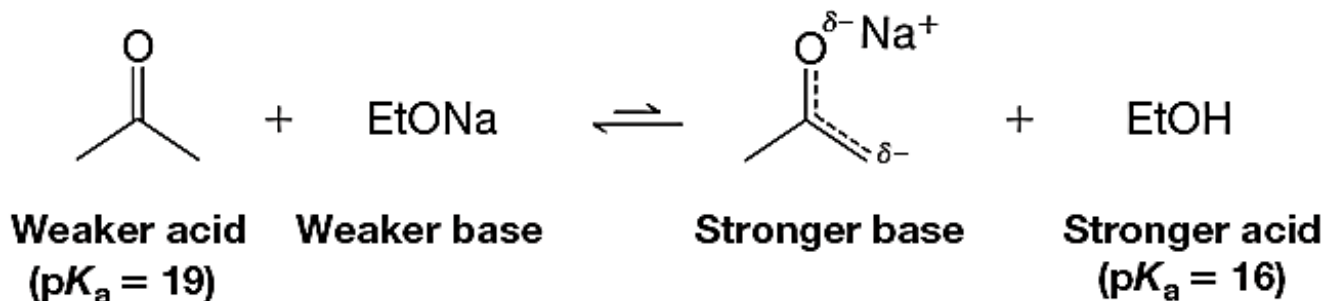
- Racemization at an α carbon takes place in the presence of acids or bases



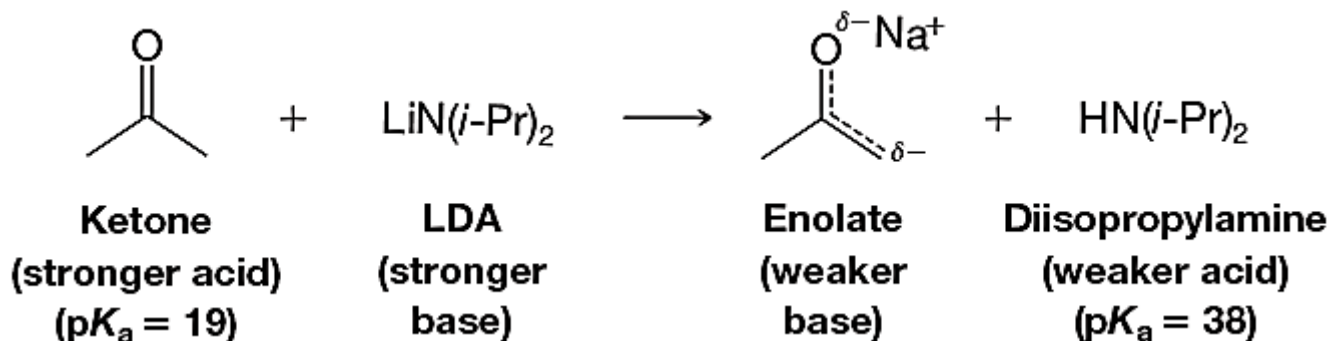
Exercise: Provide mechanism for base and acid-catalyzed enolization of the above reaction

Lithium Enolates

- If the base used is a weaker base than the enolate, then the equilibrium lies to the left

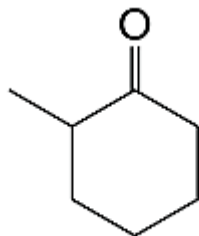


- When a very strong base such as **lithium diisopropylamide (LDA)** or $\text{LiN}(i\text{-Pr})_2$ is used, the equilibrium lies far to the right



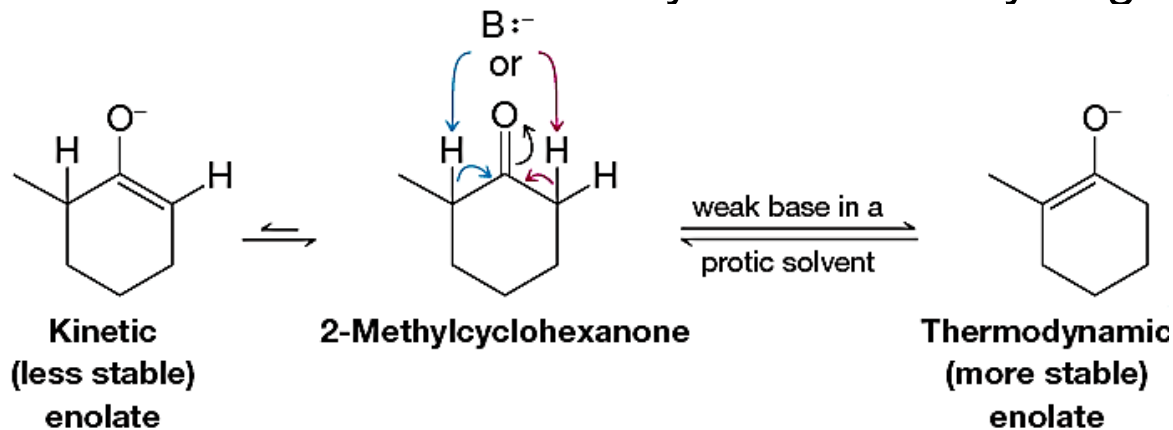
Regioselective Formation of Enolates

- An unsymmetrical ketone such as 2-methylcyclohexanone can form two possible enolates

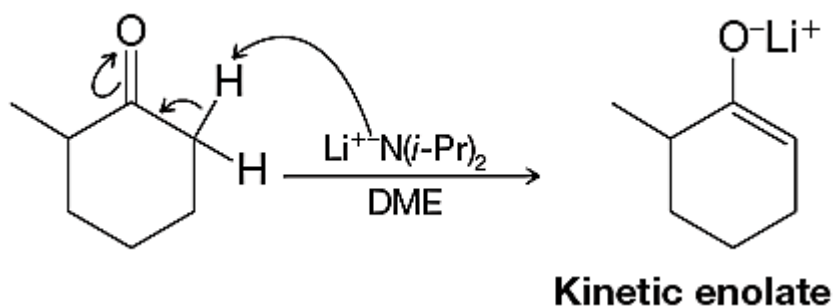


2-Methylcyclohexanone

- The **thermodynamic enolate** is that which is most stable among the possible enolates
- The **kinetic enolate** is that which is formed fastest and it is usually formed by removal of the least sterically hindered α hydrogen

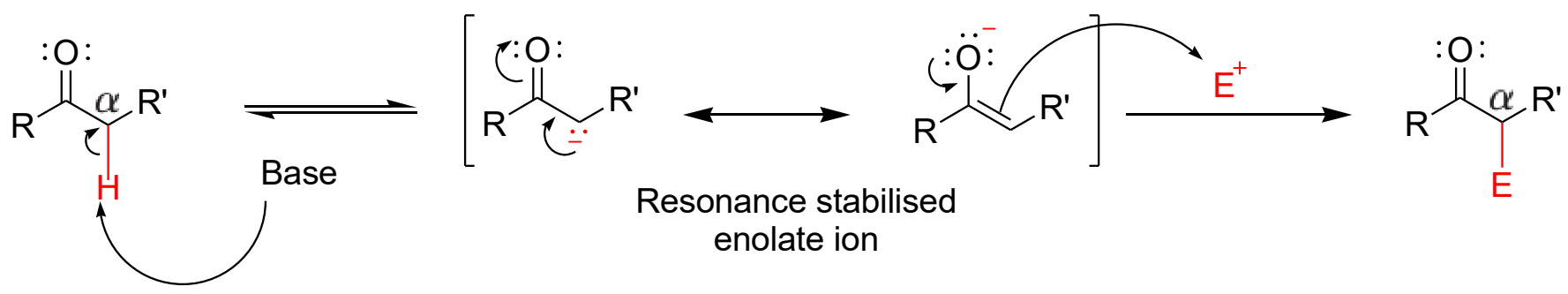


- Very strong and sterically hindered bases in an aprotic solvent, such as LDA in tetrahydrofuran (THF) or dimethoxyethane (DME) favors formation of the kinetic enolate



α -Substitution reactions of the carbonyl compounds

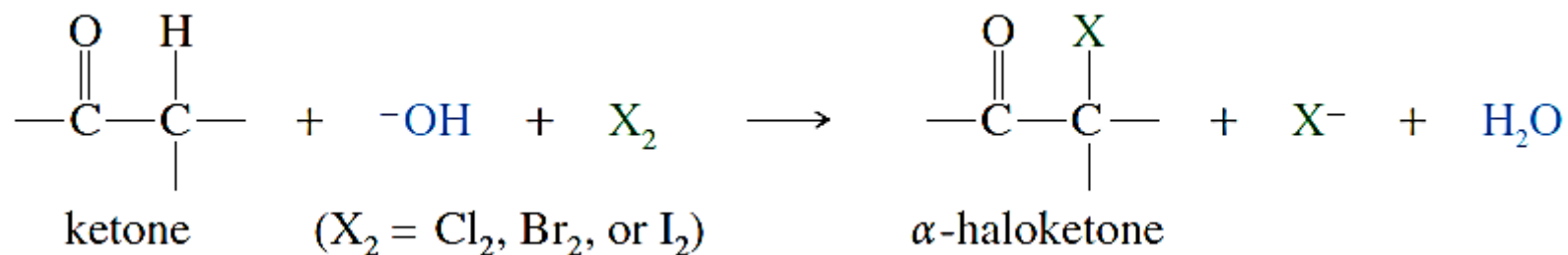
General mechanism:



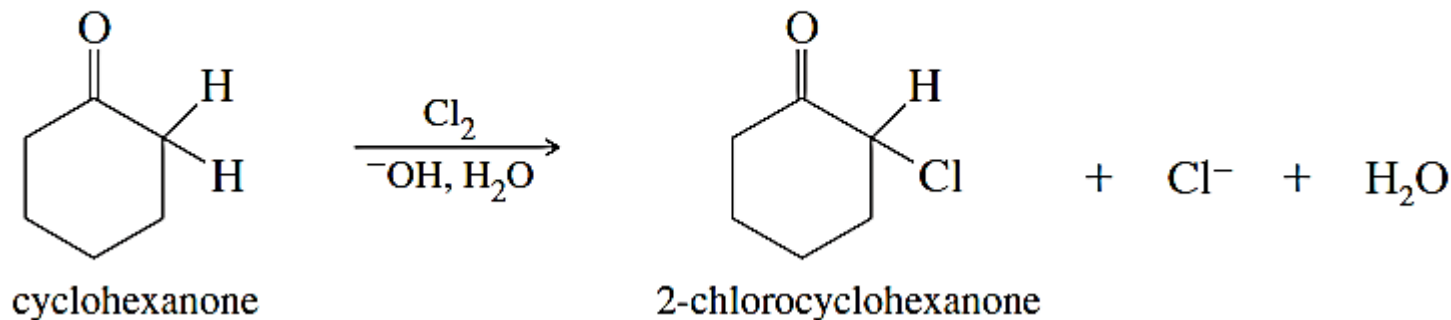
α -Carbon halogenations

(i) Base-promoted

- When a ketone is treated with a halogen and base, an α -halogenation reaction occurs

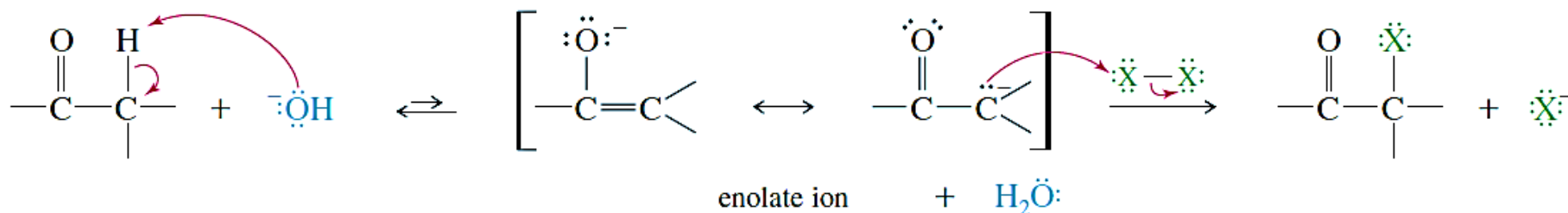


Example

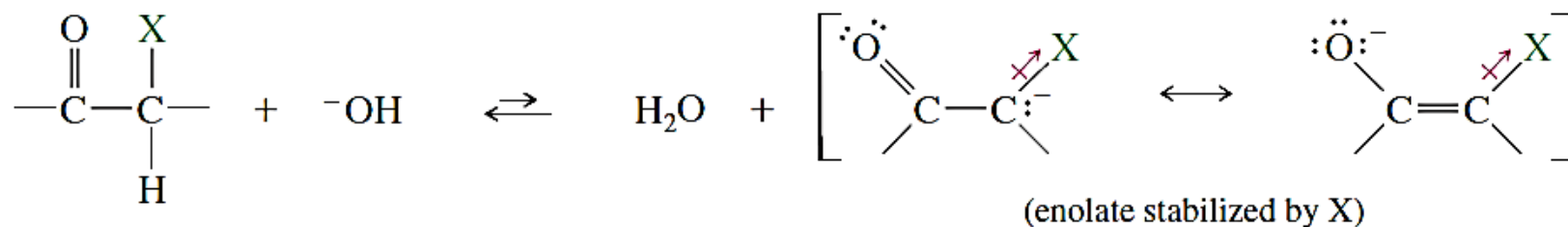


Mechanism: Base-Promoted Halogenation

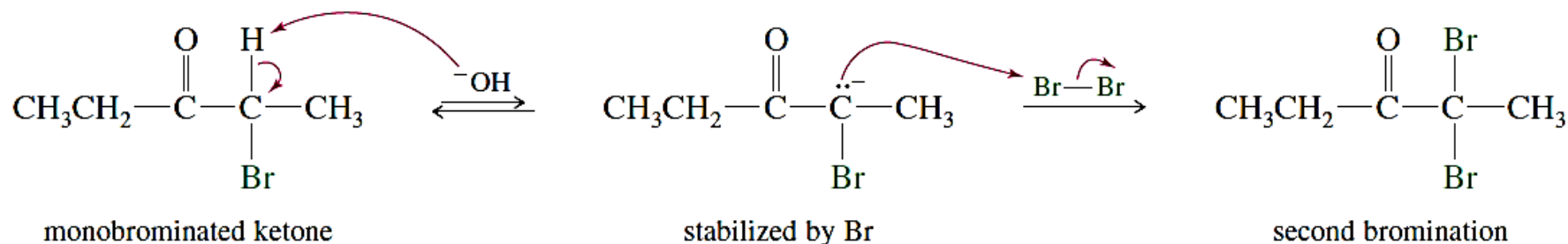
- Deprotonation of the α -carbon forms the enolate ion.
- The enolate ion attacks the electrophilic halogen



- In many cases, base-promoted halogenation leads to **multiple halogenation** because the product (the α -haloketone) is more reactive toward further halogenation than is the starting material
✓ the electron withdrawing halogen stabilizes the enolate ion.



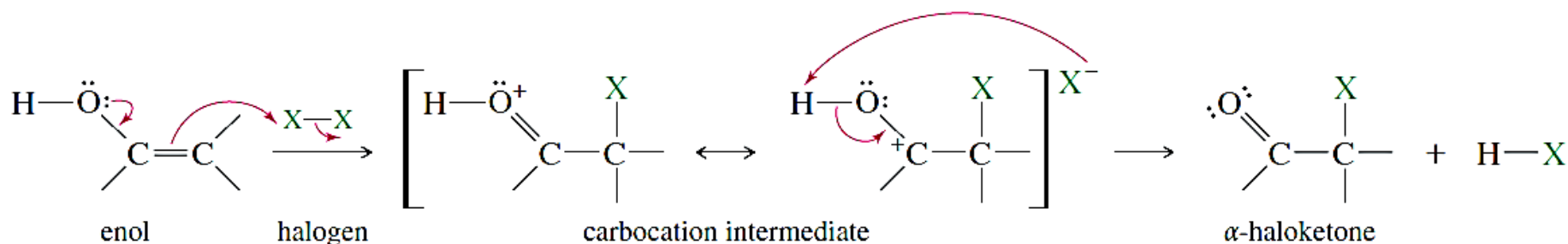
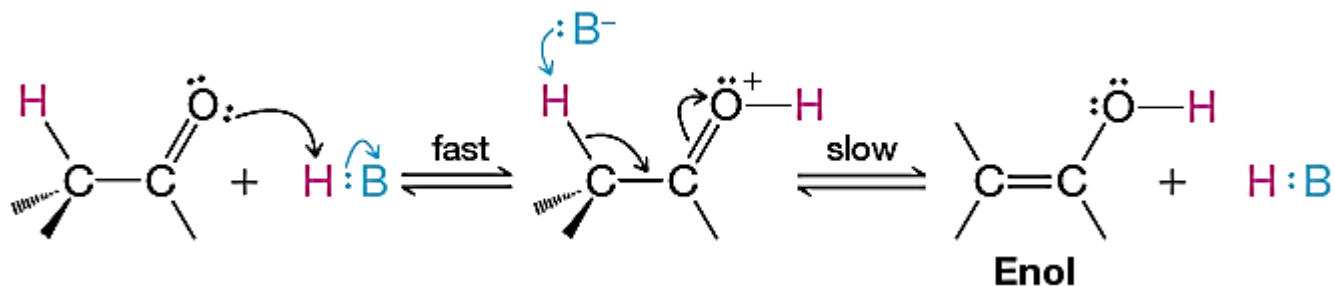
Example: bromination of pentan-3-one gives mostly 2,2-dibromopentan-3-one



Exercise: Propose a mechanism to show how acetophenone undergoes base promoted chlorination to give trichloroacetophenone.

(ii) Acid-Catalyzed

- Acidic halogenation can selectively replace just one hydrogen or more than one, depending on the amount of the halogen added



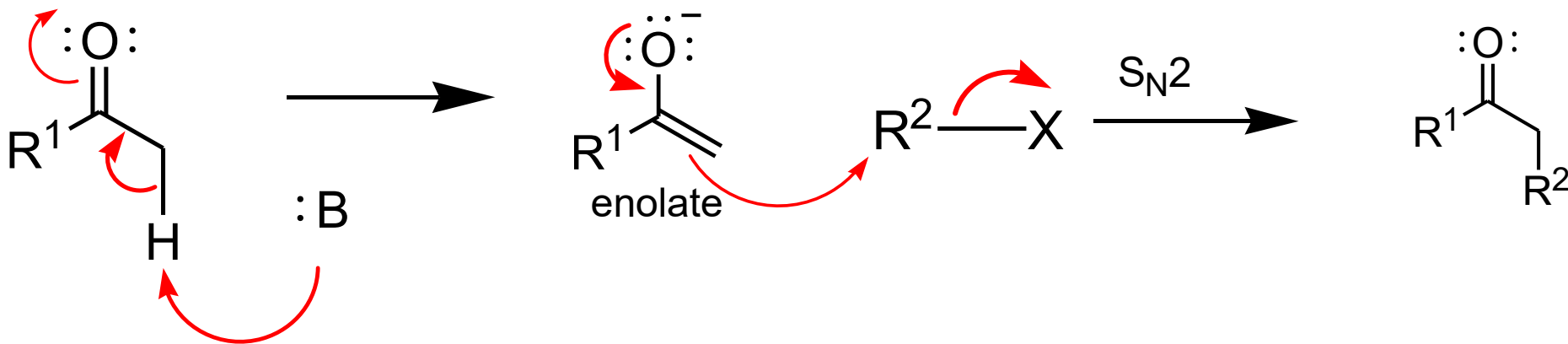
Exercise: Propose a mechanism for the acid-catalyzed monobromination of pentan-3-one.

α -Carbon alkylations

Important considerations that affect all alkylations

Two steps involved in the alkylation reactions:

1. The formation of a stabilized anion (usually an enolate) by deprotonation with **strong** base
2. The substitution reaction: attack of the nucleophilic anion on an electrophile (**e.g. Alkyl halide**).

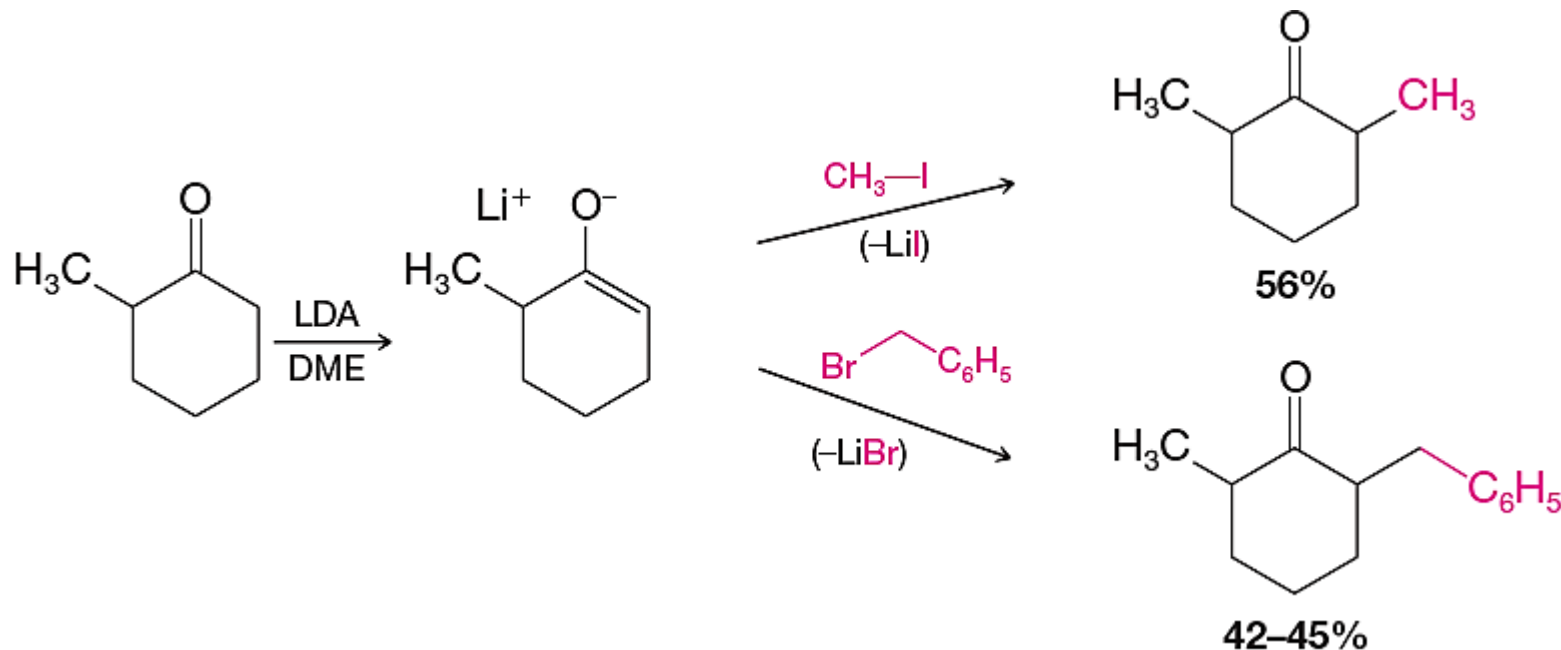


Two approaches on the choice of a base

1. Deprotonate with strong base (with a conjugate acid of pK_a greater than that of the carbonyl compound) can be chosen followed by addition of the electrophile.
2. Use a weaker base in the presence of the electrophile

Direct Alkylation of Ketones via Lithium Enolates

- Lithium enolates enhance regioselective alkylation of ketones



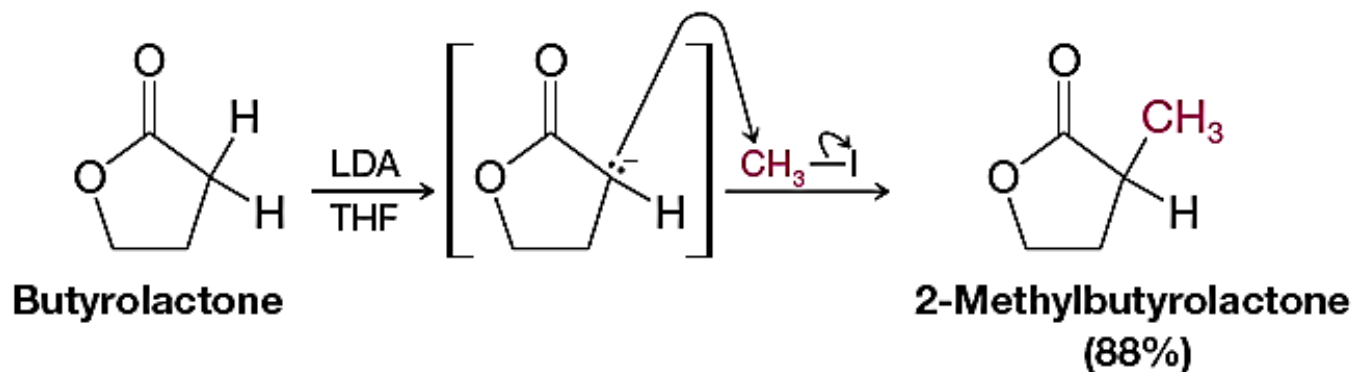
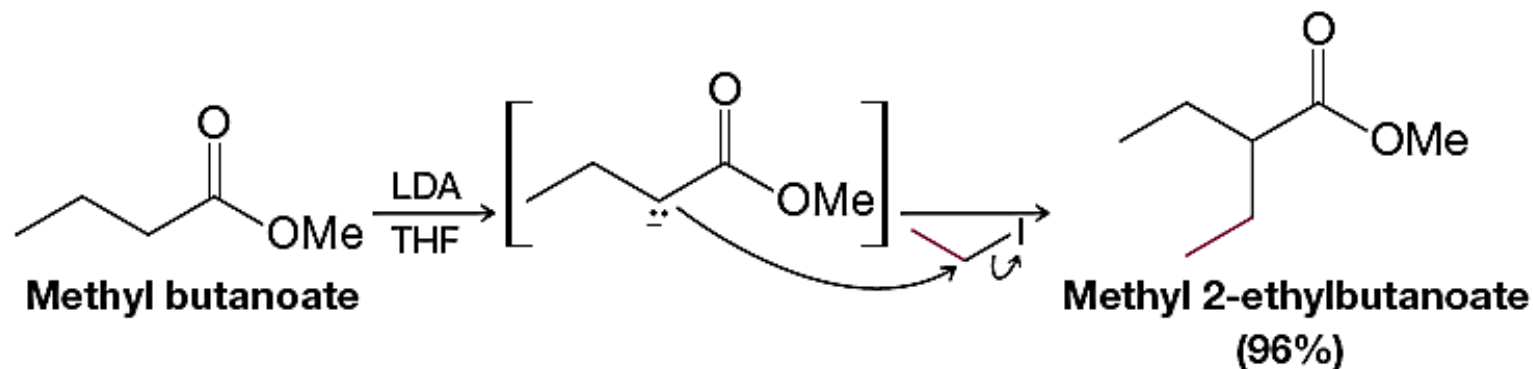
- These alkylations are $\text{S}_{\text{N}}2$ reactions

Limitations:

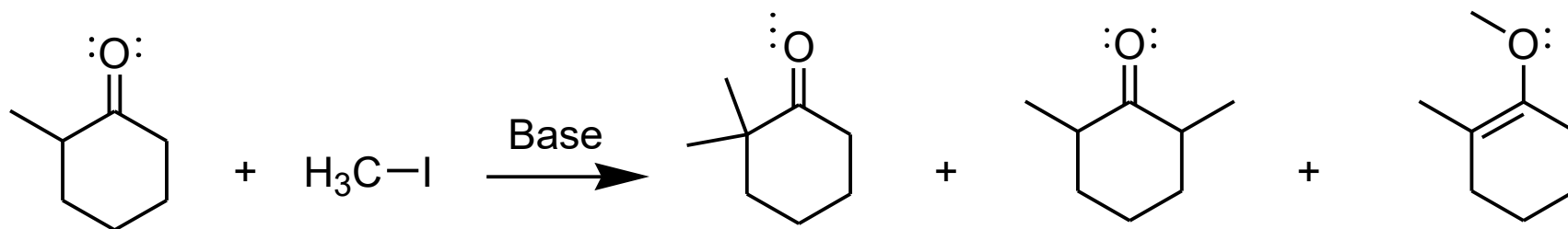
- Only works with primary alkyl, primary benzylic, and primary allylic halides
- Elimination occurs when secondary and all tertiary halides are used

Direct Alkylation of Esters

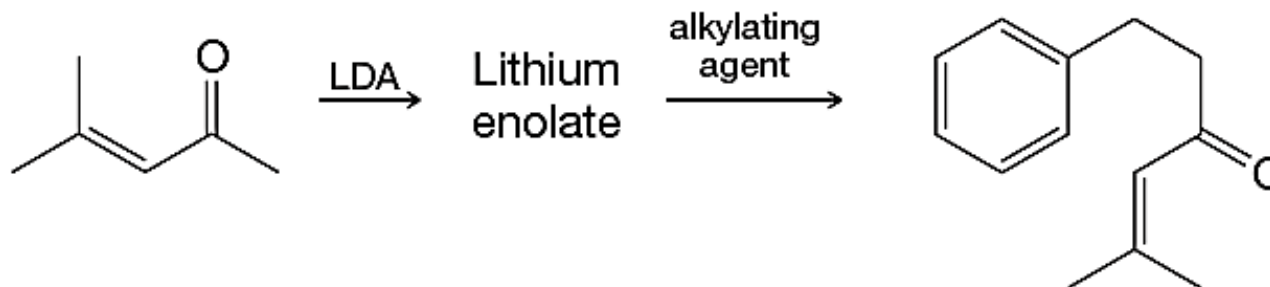
- Esters can be alkylated as shown below



Provide mechanism for the formation of each of the products below.

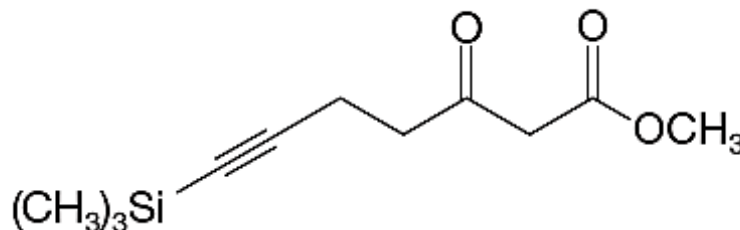


Give the structures of the enolate and the alkylating agent in the following reaction.



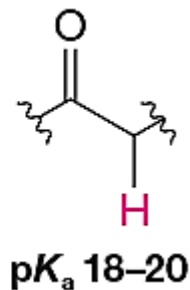
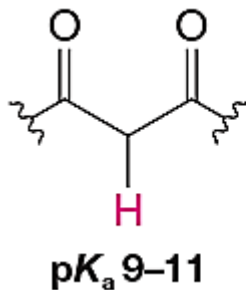
Dienolates can be formed from β -keto esters using two equivalents of LDA.

The more basic enolate of the dienolate can be alkylated selectively. Write a reaction for synthesis of the following compound using a dienolate and the appropriate alkyl halide

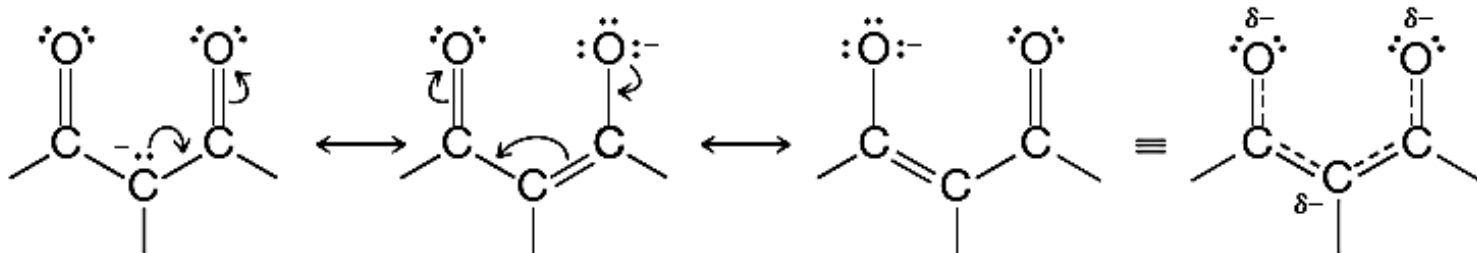
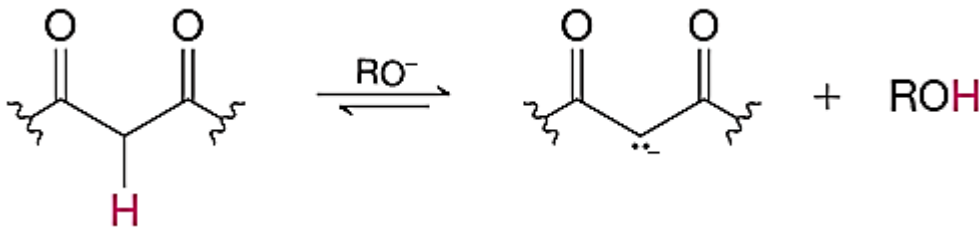


Enolates of β -Dicarbonyl Compounds

- Hydrogen atoms that are between two carbonyl groups are more acidic than α hydrogens adjacent to only one carbonyl group



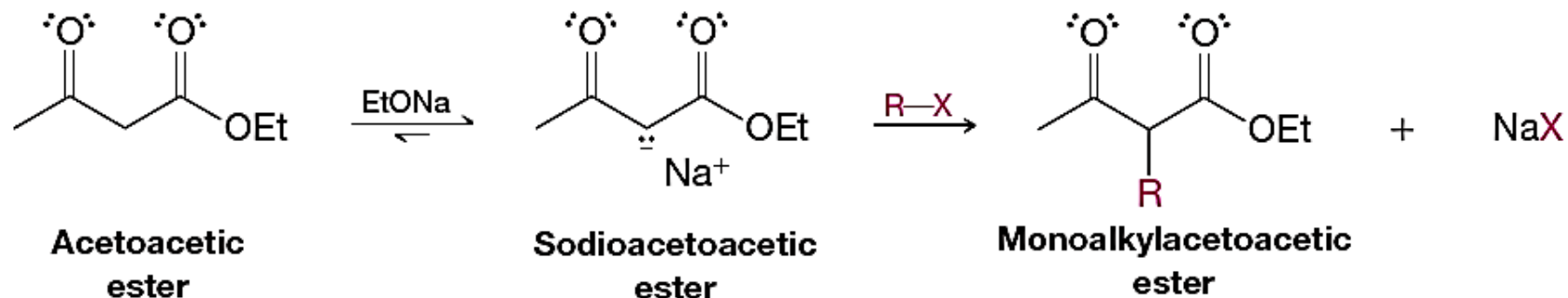
- A weaker base than LDA, such as an alkoxide, can be used to form an enolate from a β -dicarbonyl compound



Contributing resonance structures

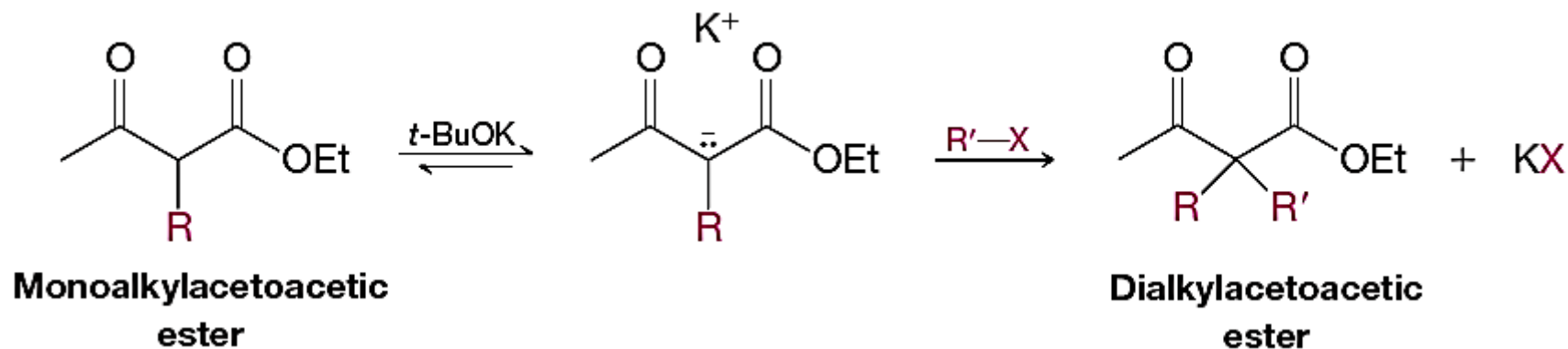
Resonance hybrid

Synthesis of Methyl Ketones from β -Dicarbonyl Compounds: The Acetoacetic Ester Synthesis



Dialkylation:

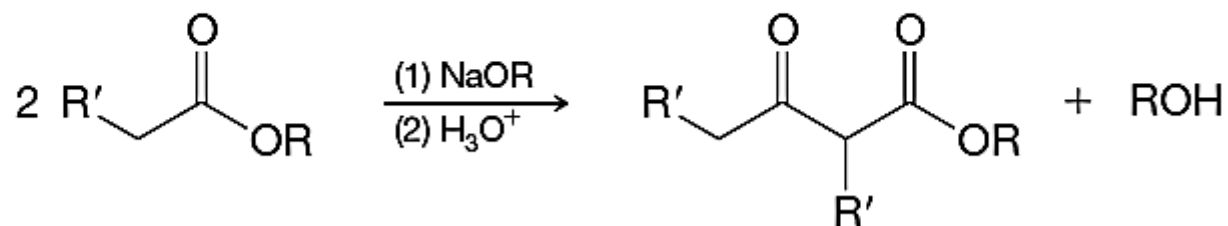
- Second alkylation requires use of a stronger base than ethoxide ion. Potassium *tert*-butoxide is commonly used



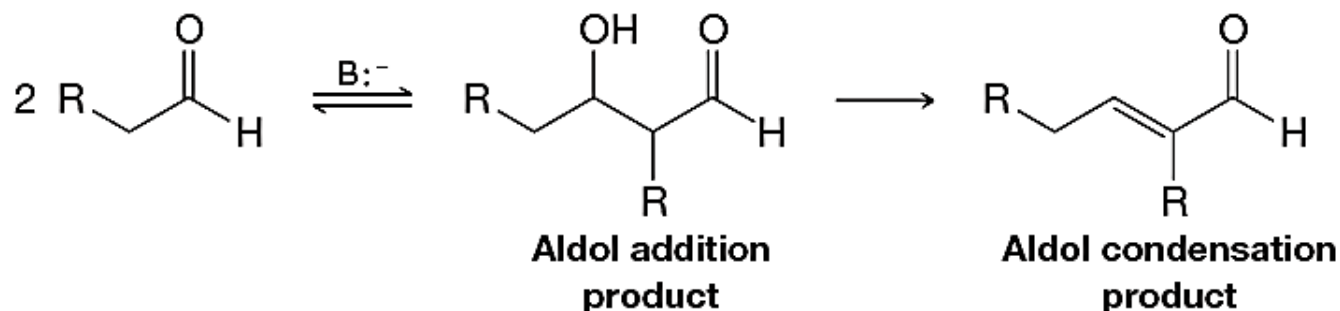
**Condensation and Conjugate Addition Reactions of Carbonyl
Compounds (Ch.19, Solomons)**

- **Carbonyl condensation:** the enolate or enol of one carbonyl compound reacts with the carbonyl group of another
- The main types of condensation reactions are the **Claisen condensation** and the **aldol condensation**

An Example of a Claisen Condensation

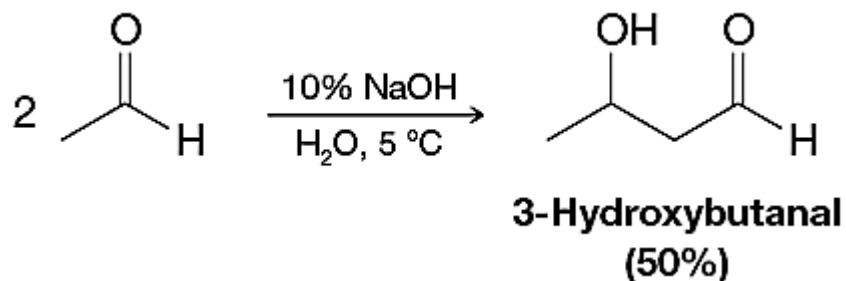


An Example of an Aldol Addition and Condensation



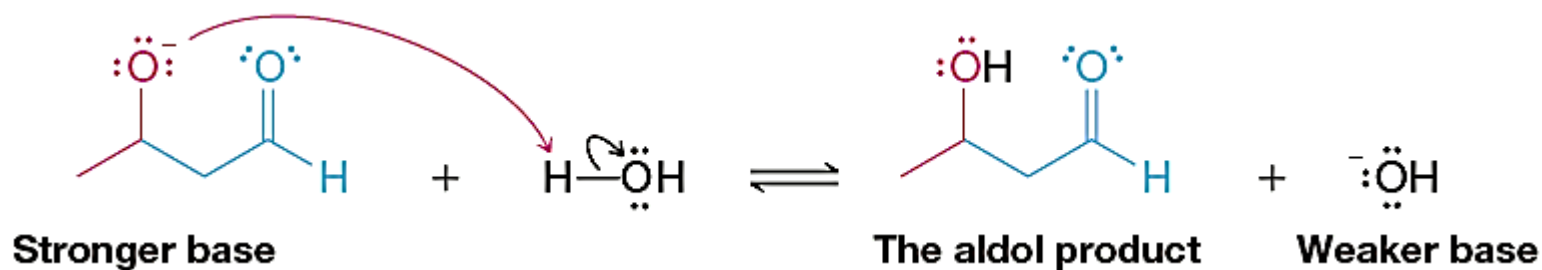
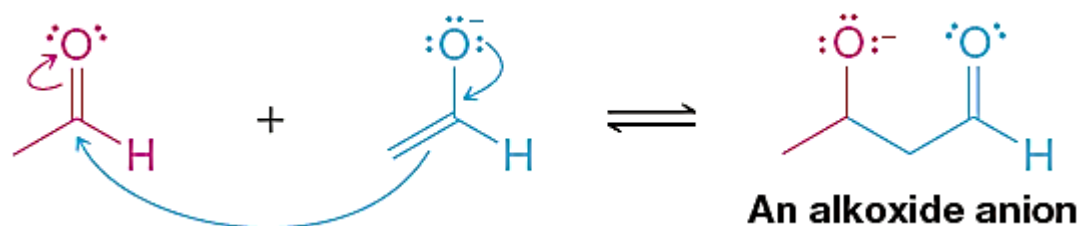
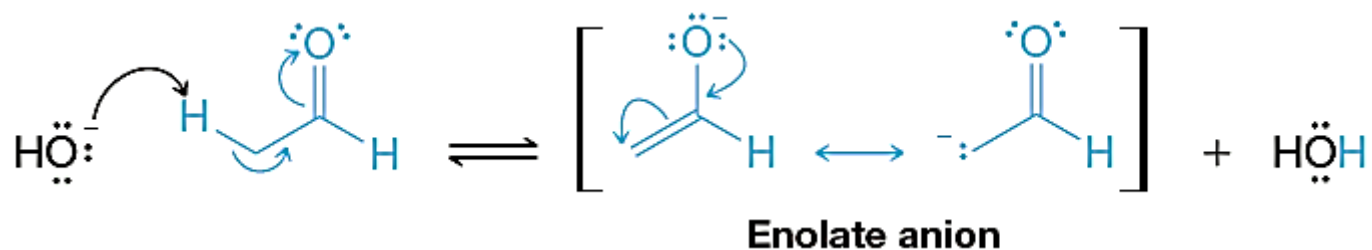
Aldol Reactions: Addition of Enolates and Enols to Aldehydes and Ketones

- An important class method of C–C bond-forming reactions
- It begins with addition of an enolate or enol to the carbonyl group of an aldehyde or ketone
- This produce β -hydroxy aldehyde or ketone as the initial product



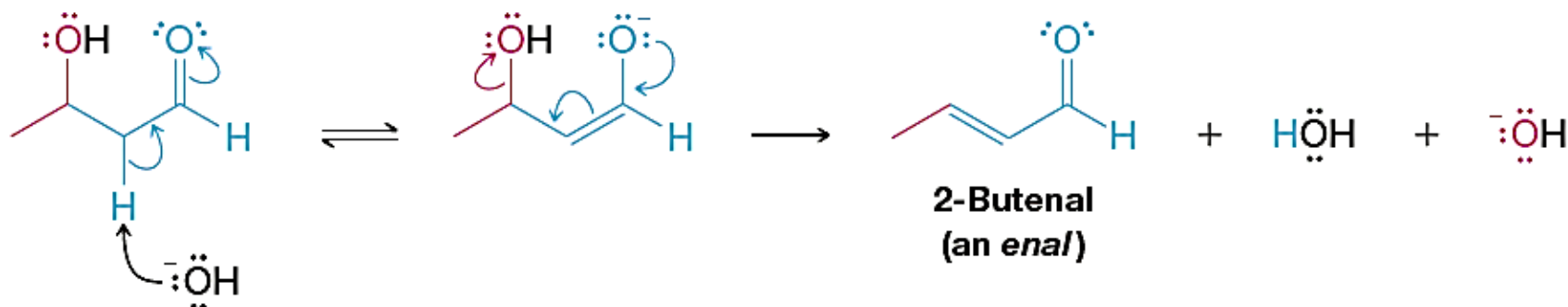
- “**Aldol**” because product contains both an **aldehyde** and an **alcohol** functional group

A Mechanism for the Reaction: Addition



Aldol Condensation Reactions: Dehydration of the Aldol Addition Product

- Dehydrating an aldol addition product gives a conjugated α,β -unsaturated carbonyl system
- The dehydration equilibrium is irreversible

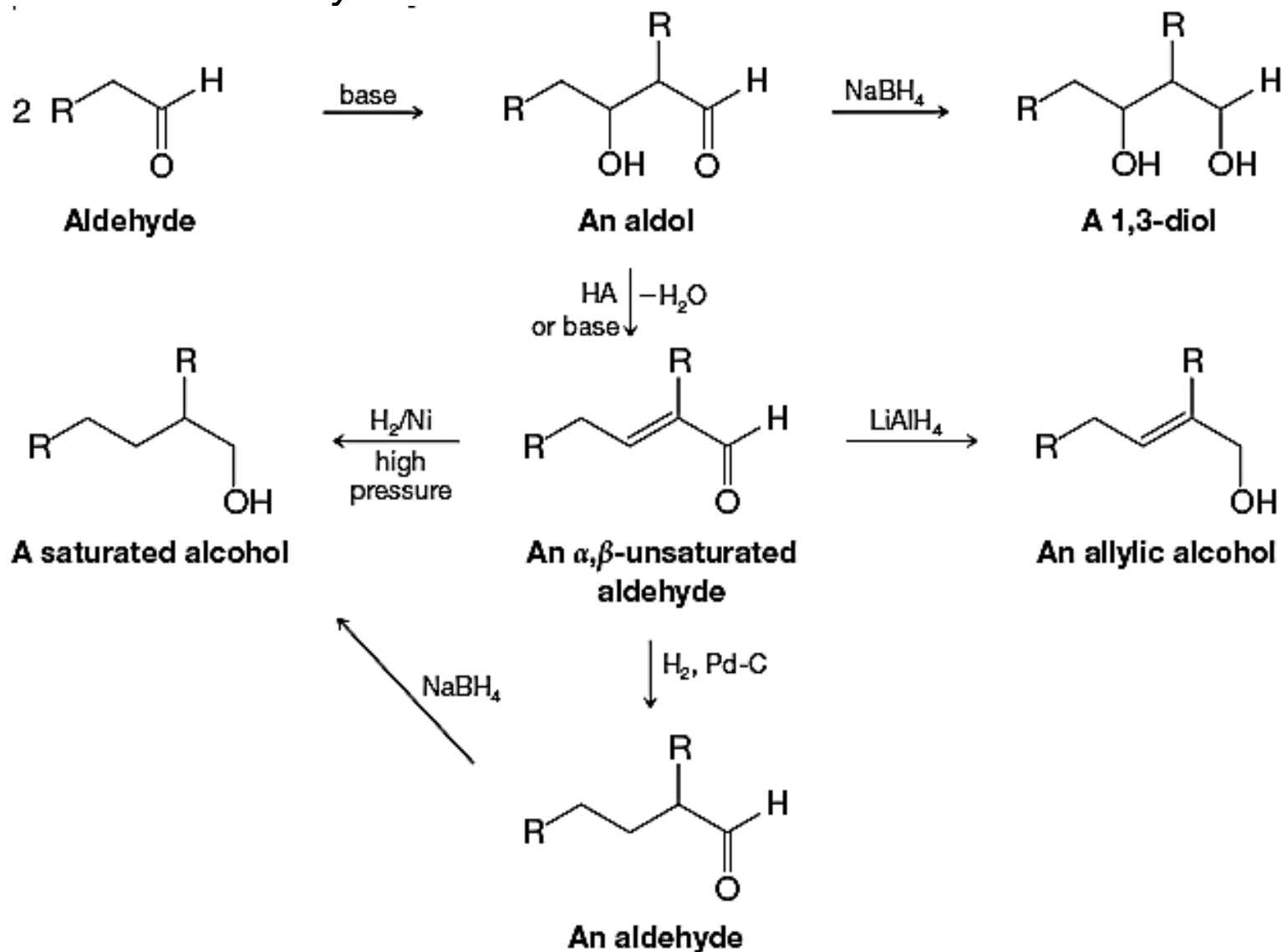


- The conjugation between the double bonds of the alkene and carbonyl groups stabilize the product

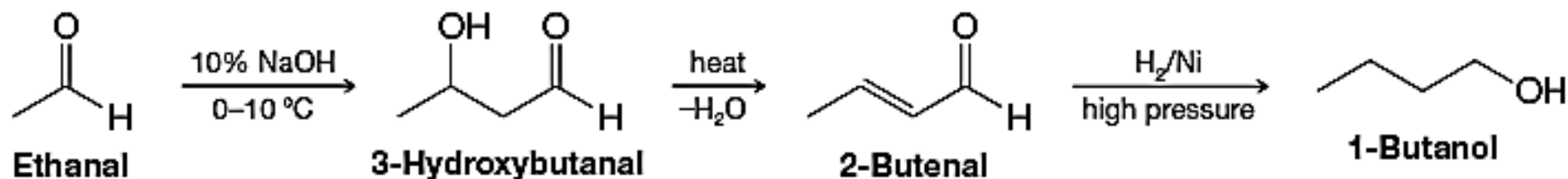
Do the acid-catalyzed aldol condensation mechanism (Addition & dehydration)

Synthetic Applications of Aldol Reactions

The Aldol Reaction in Synthesis

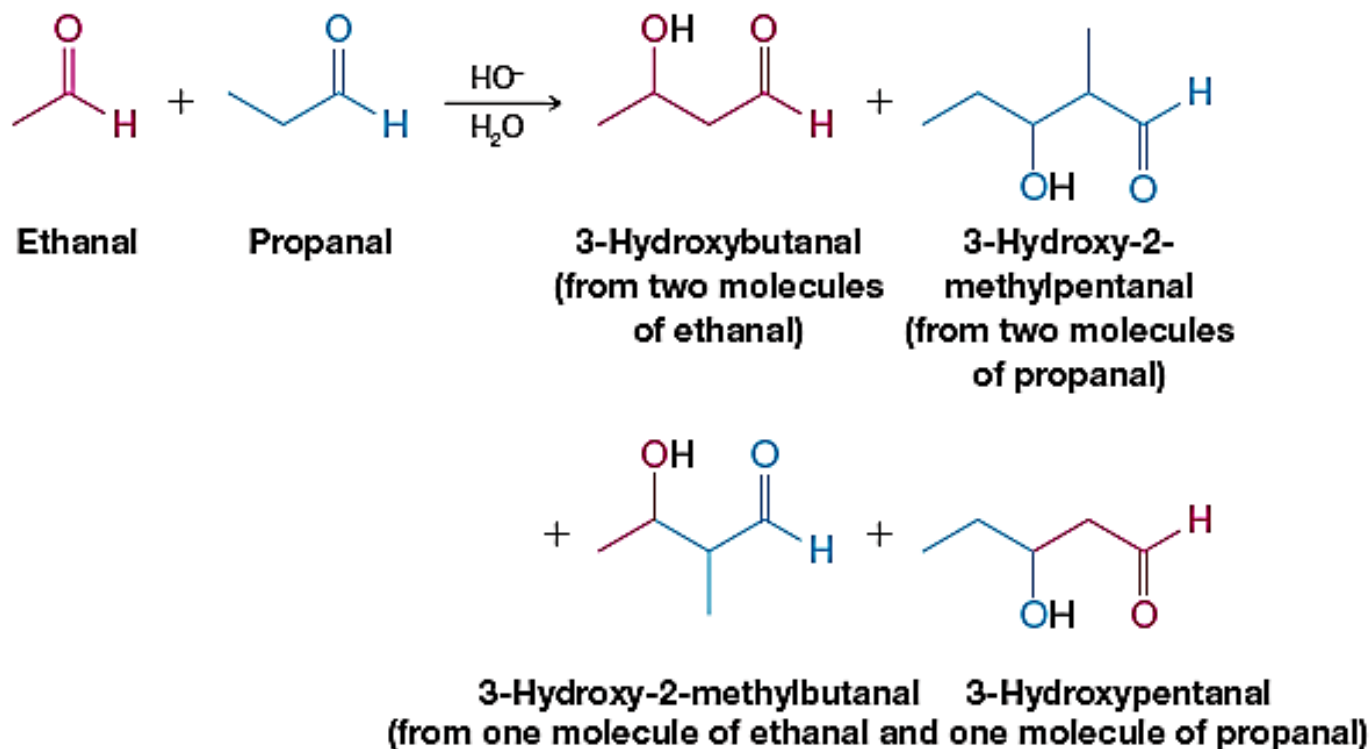


Industrial process for the synthesis of 1-butanol from ethanal.



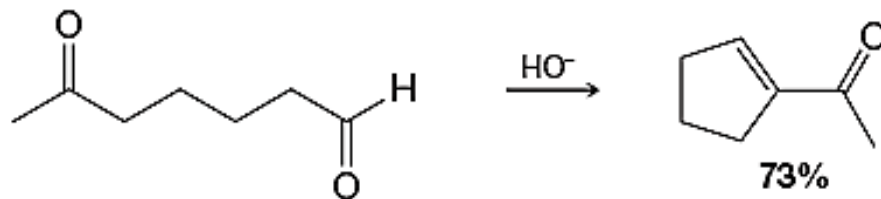
Crossed Aldol Condensations

- An **aldol reaction** from two different carbonyl compounds
- Can lead to a mixture of products from various pairings of the carbonyl reactants



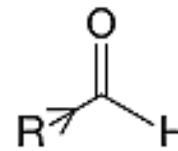
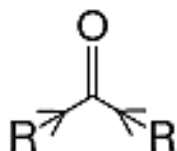
Cyclizations via Aldol Condensations

- molecules with five and six-membered rings (and sometimes even larger rings) can be built by intramolecular aldol condensation



Propose mechanism for this Aldol cyclization and write the structures of the two other products that might have resulted

- At least three different enolates
- The thermodynamic enolate is favored over kinetic enolate
- The aldehyde group undergoes addition preferentially due to its greater reactivity toward nucleophilic additions



What starting compound would you use in an aldol cyclization to prepare each of the following?

